STAT GU4204 Section 001 — Final Exam

Variant E: Hybrid / Most-Likely — SOLUTIONS

Spring 2026 Total: 100 points

Part A: Short Conceptual Questions

Question 1 (4 points) Sufficiency and the Factorization Criterion

Definition. A statistic $T = r(\mathbf{X})$ is *sufficient* for θ if the conditional distribution of $\mathbf{X}$ given $T = t$ does not depend on θ. Intuitively, once you know T, the data $\mathbf{X}$ contain no further information about θ.

Factorization Criterion (Fisher–Neyman). $T = r(\mathbf{X})$ is sufficient for θ if and only if the joint density (or pmf) of $\mathbf{X}$ factors as

$$f_n(\mathbf{x};\theta) = u(\mathbf{x}) \cdot \nu\big(r(\mathbf{x}),\, \theta\big),$$

where $u(\mathbf{x}) \geq 0$ does not depend on θ and $\nu(t,\theta)$ depends on $\mathbf{x}$ only through $r(\mathbf{x})$.

Bernoulli example. For $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$:

$$f_n(\mathbf{x};p) = \prod_{i=1}^{n} p^{x_i}(1-p)^{1-x_i} = p^{\sum x_i}(1-p)^{n-\sum x_i} = \underbrace{1}_{u(\mathbf{x})} \cdot \underbrace{p^t(1-p)^{n-t}}_{\nu(t,p)},$$

where $t = \sum_{i=1}^n x_i$. Hence by the factorization criterion, $T = \sum_{i=1}^n X_i$ is $\boxed{\text{sufficient for } p}$.

Question 2 (3 points) Cramér-Rao Lower Bound

CRLB statement. Let $T = T(\mathbf{X})$ be an unbiased estimator of $g(\theta)$ based on a random sample $X_1, \ldots, X_n$. Under regularity conditions (A.1)–(A.3),

$$\boxed{\operatorname{Var}_\theta(T) \geq \frac{[g'(\theta)]^2}{n\, I(\theta)}\,,}$$

where $I(\theta) = E_\theta[\dot{\ell}(X;\theta)^2]$ is the Fisher information for one observation. For $g(\theta) = \theta$ this reduces to $\operatorname{Var}_\theta(T) \geq 1/(nI(\theta))$.

Uniform$(0,\theta)$**.** The density is $f(x;\theta) = \frac{1}{\theta}\mathbb{1}\{0 < x < \theta\}$, whose support $\{x : f(x;\theta) > 0\} = (0,\theta)$ *depends on* θ. This violates condition $\boxed{(A.1)}$ (support independent of θ). Without (A.1) we cannot interchange differentiation and integration in the proof of the CRLB, so the bound is not directly applicable. (Indeed, the MLE $X_{(n)}$ has variance shrinking like $1/n^2$, which would violate any $1/n$ bound.)

Question 3 (3 points) **Neyman-Pearson Lemma**

Statement. Let $X_1, \ldots, X_n$ be a random sample from $f(\cdot;\theta)$ where $\Omega = \{\theta_0, \theta_1\}$. Consider testing the simple hypotheses

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_1 : \theta = \theta_1.$$

Let $f_0(\mathbf{x})$ and $f_1(\mathbf{x})$ be the joint densities under θ_0 and θ_1 respectively. For any $k > 0$, define the test δ^* that

- rejects H_0 if $f_1(\mathbf{x}) > k\, f_0(\mathbf{x})$,
- does not reject H_0 if $f_1(\mathbf{x}) < k\, f_0(\mathbf{x})$,
- is arbitrary on the boundary set.

Then for any other test δ with $\alpha(\delta) \le \alpha(\delta^*)$, we have $\beta(\delta) \ge \beta(\delta^*)$. That is, $\boxed{\delta^* \text{ is the most powerful (MP) test of it}}$ among all tests with Type I error $\le \alpha(\delta^*)$, the likelihood ratio test minimizes Type II error (equivalently, maximizes power at θ_1).

Part B: Mid-Length Problems

Question 4 (15 points) **MLE, Sufficiency, and Fisher Information for the Exponential**

(a) MLE of θ. The likelihood is

$$L(\theta) = \prod_{i=1}^{n} \theta\, e^{-\theta x_i} = \theta^n \exp\left(-\theta \sum_{i=1}^{n} x_i\right).$$

The log-likelihood is

$$\ell(\theta) = n \log \theta - \theta \sum_{i=1}^{n} x_i.$$

Differentiating and setting equal to zero:

$$\ell'(\theta) = \frac{n}{\theta} - \sum_{i=1}^{n} x_i = 0 \implies \theta = \frac{n}{\sum_{i=1}^{n} x_i} = \frac{1}{\overline{X}_n}.$$

The second derivative is $\ell''(\theta) = -n/\theta^2 < 0$, confirming a maximum. Hence

$$\boxed{\hat{\theta}_n = \frac{1}{\overline{X}_n} = \frac{n}{\sum_{i=1}^{n} X_i}}.$$

(b) Sufficiency of $T = \sum X_i$. The joint density is

$$f_n(\mathbf{x};\theta) = \theta^n \exp\left(-\theta \sum_{i=1}^{n} x_i\right) \prod_{i=1}^{n} \mathbb{1}\{x_i > 0\} = \underbrace{\prod_{i=1}^{n} \mathbb{1}\{x_i > 0\}}_{u(\mathbf{x})} \cdot \underbrace{\theta^n e^{-\theta t}}_{\nu(t,\theta)},$$

where $t = \sum_{i=1}^n x_i$. The function $u(\mathbf{x})$ does not involve θ, and $\nu(t, \theta)$ depends on $\mathbf{x}$ only through t. By the Fisher–Neyman factorization criterion, $T = \sum_{i=1}^n X_i$ is $\boxed{\text{sufficient for } \theta}$.

(c) Fisher information and efficiency. The score for one observation is

$$\dot{\ell}(x; \theta) = \frac{\partial}{\partial \theta}\big[\log \theta - \theta x\big] = \frac{1}{\theta} - x.$$

Then

$$I(\theta) = E_\theta[\dot{\ell}(X; \theta)^2] = \mathrm{Var}_\theta\left(\frac{1}{\theta} - X\right) = \mathrm{Var}_\theta(X) = \frac{1}{\theta^2}.$$

(Equivalently, $-E[\ddot{\ell}] = -E[-1/\theta^2] = 1/\theta^2$.) For the full sample, $I_n(\theta) = nI(\theta) = \boxed{n/\theta^2}$.

Efficiency. First check unbiasedness. Since $T = \sum X_i \sim \mathrm{Gamma}(n, \theta)$,

$$E[\hat{\theta}_n] = E\left[\frac{n}{T}\right] = n \int_0^\infty \frac{1}{t} \cdot \frac{\theta^n}{\Gamma(n)} t^{n-1} e^{-\theta t}\, dt = \frac{n\theta^n}{\Gamma(n)} \cdot \frac{\Gamma(n-1)}{\theta^{n-1}} = \frac{n\theta}{n-1}.$$

Hence $E[\hat{\theta}_n] = \frac{n}{n-1}\theta \neq \theta$ for finite n — $\hat{\theta}_n$ is *biased*. Therefore the CRLB (which applies to *unbiased* estimators of θ) does not directly compare to $\mathrm{Var}(\hat{\theta}_n)$, and we conclude $\hat{\theta}_n$ is $\boxed{\text{not an efficient unbiased estimator of } \theta}$ (The bias-corrected estimator $\frac{n-1}{n}\hat{\theta}_n$ is unbiased but still does not attain the CRLB exactly.)

(d) Asymptotic distribution. By the asymptotic normality of the MLE under regularity conditions,

$$\boxed{\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N\left(0, \frac{1}{I(\theta)}\right) = N(0, \theta^2).}$$

(One can verify this directly via the delta method: $\sqrt{n}(\overline{X}_n - 1/\theta) \to N(0, 1/\theta^2)$ and $g(x) = 1/x$ has $g'(1/\theta) = -\theta^2$, so $\sqrt{n}(1/\overline{X}_n - \theta) \to N(0, (1/\theta^2) \cdot \theta^4) = N(0, \theta^2)$.)

Question 5 (15 points) Bayesian Inference for a Bernoulli Proportion

(a) Posterior derivation. The prior is

$$\xi(p) = \frac{1}{B(\alpha, \beta)} p^{\alpha-1}(1-p)^{\beta-1}, \quad p \in (0, 1).$$

The likelihood (given $\sum X_i = s$) is

$$f_n(\mathbf{x} \mid p) = p^s (1-p)^{n-s}.$$

By Bayes' rule,

$$\xi(p \mid \mathbf{x}) \propto f_n(\mathbf{x} \mid p)\, \xi(p) \propto p^s(1-p)^{n-s} \cdot p^{\alpha-1}(1-p)^{\beta-1} = p^{(\alpha+s)-1}(1-p)^{(\beta+n-s)-1}.$$

This is the kernel of a Beta distribution. Therefore

$$\boxed{p \mid \mathbf{X} \sim \mathrm{Beta}(\alpha + \textstyle\sum_{i=1}^n X_i,\ \beta + n - \sum_{i=1}^n X_i)\,.}$$

(b) Bayes estimator under squared error loss. Under squared-error loss, the Bayes estimator is the posterior mean. For $\mathrm{Beta}(\alpha', \beta')$, the mean is $\alpha'/(\alpha' + \beta')$. Thus

$$\hat{p}_B = E[p \mid \mathbf{X}] = \frac{\alpha + \sum X_i}{\alpha + \beta + n}.$$

Writing $w_n = n/(n+\alpha+\beta)$,

$$\hat{p}_B = \frac{\alpha+\beta}{n+\alpha+\beta} \cdot \underbrace{\frac{\alpha}{\alpha+\beta}}_{\text{prior mean}} + \frac{n}{n+\alpha+\beta} \cdot \underbrace{\frac{\sum X_i}{n}}_{\hat{p}_n = \overline{X}_n}.$$

$$\boxed{\hat{p}_B = (1-w_n) \cdot \frac{\alpha}{\alpha+\beta} + w_n \cdot \overline{X}_n, \quad w_n = \frac{n}{n+\alpha+\beta}.}$$

(c) Numerical values. With $\alpha = \beta = 1$, $n = 10$, $\sum X_i = 7$:

$$\hat{p}_B = \frac{1+7}{1+1+10} = \frac{8}{12} = \boxed{\tfrac{2}{3} \approx 0.667}, \qquad \hat{p}_n = \frac{7}{10} = \boxed{0.7}.$$

The Bayes estimator pulls the MLE slightly toward the prior mean of 1/2 (the uniform prior centers at 1/2), giving a more conservative estimate.

(d) Asymptotic relationship. As $n \to \infty$ with α, β fixed, $w_n = n/(n+\alpha+\beta) \to 1$, so

$$\boxed{\hat{p}_B = \hat{p}_n + O_p(1/n) \implies \hat{p}_B \xrightarrow{P} p \text{ and } \sqrt{n}(\hat{p}_B - \hat{p}_n) \xrightarrow{P} 0.}$$

Why: the data swamp the prior — the posterior concentrates around the MLE (the data's information grows linearly in n, while the prior contributes a fixed amount equivalent to $\alpha+\beta$ "pseudo-observations"). This is the standard Bernstein–von Mises phenomenon.

Part C: Major Problems

Question 6 (20 points) t-Test, Confidence Interval, and Test/CI Duality

Given: $n = 16$, $\overline{X}_n = 12.5$, $s_n^2 = 4.0$ (so $s_n = 2$), σ unknown.

(a) Two-sided t-test of $\mu = 11$ at $\alpha_0 = 0.05$.

Test statistic:

$$U_n = \frac{\sqrt{n}(\overline{X}_n - \mu_0)}{s_n} = \frac{\sqrt{16}(12.5-11)}{2} = \frac{4 \cdot 1.5}{2} = \boxed{3.0}.$$

Under H_0, $U_n \sim t_{n-1} = t_{15}$. Reject H_0 if $|U_n| \geq t_{0.025,15} = 2.131$.

Since $|U_n| = 3.0 > 2.131$, $\boxed{\text{reject } H_0}$: there is evidence at the 5% level that $\mu \neq 11$.

(b) 95% CI for μ. Using the pivot $\sqrt{n}(\overline{X}_n - \mu)/s_n \sim t_{n-1}$:

$$\overline{X}_n \pm t_{0.025,15} \cdot \frac{s_n}{\sqrt{n}} = 12.5 \pm 2.131 \cdot \frac{2}{4} = 12.5 \pm 1.0655.$$

$$\boxed{\text{95\% CI: } (11.4345,\ 13.5655).}$$

Duality check. The value $\mu_0 = 11$ does *not* lie in $(11.4345, 13.5655)$. By the test/CI duality (DeGroot Theorem 10.3): the level-α_0 test of $H_0 : \mu = \mu_0$ rejects H_0 if and only if $\mu_0 \notin$ the $(1-\alpha_0)$

CI. Indeed, both procedures here reach the same conclusion (reject H_0). The CI is precisely the set of null values μ_0 that are not rejected by the level-α_0 t-test.

(c) One-sided test $H_0 : \mu \leq 11$ vs $H_1 : \mu > 11$.

Same test statistic $U_n = 3.0$, but the rejection rule is now $U_n \geq t_{0.05,15} = 1.753$. Since $3.0 > 1.753$, $\boxed{\text{reject } H_0}$: there is evidence at level 0.05 that $\mu > 11$.

(Note that the size of this test is $\sup_{\mu \leq 11} P_\mu(U_n \geq 1.753) = P_{\mu=11}(U_n \geq 1.753) = 0.05$, achieved at the boundary.)

(d) Asymptotic null distribution of $-2 \log \Lambda$. For the two-sided LRT in (a), with H_0 specifying one coordinate of $\theta = (\mu, \sigma^2)$ (the constraint $\mu = 11$, with σ^2 free), Wilks' Theorem gives

$$\boxed{-2 \log \Lambda(\mathbf{X}) \xrightarrow{d} \chi^2_1 \text{ under } H_0.}$$

The degrees of freedom equal $\dim \Omega - \dim \Omega_0 = 2 - 1 = 1$, matching the single linear constraint imposed by H_0.

Question 7 (20 points) Neyman-Pearson Test, Type I/II Errors, and UMP

Setup: $X_1, \ldots, X_n$ iid $N(\theta, 1)$, $n = 25$. Test $H_0 : \theta = 0$ vs $H_1 : \theta = 0.5$.

(a) Most powerful test via Neyman-Pearson. The likelihood ratio is

$$\frac{f_1(\mathbf{x})}{f_0(\mathbf{x})} = \exp\Big(-\tfrac{1}{2}\sum(x_i - 0.5)^2 + \tfrac{1}{2}\sum x_i^2\Big) = \exp\Big(0.5\sum x_i - \tfrac{n}{2}(0.5)^2\Big) = \exp\Big(0.5\sum x_i - \tfrac{n}{8}\Big).$$

By the Neyman-Pearson Lemma, the MP test rejects H_0 when this ratio exceeds k, equivalently when

$$0.5\sum x_i - n/8 > \log k \iff \sum x_i > 2(\log k + n/8) \iff \overline{X}_n > c,$$

for some threshold c. Hence the rejection region has the form $\boxed{\{\overline{X}_n > c\}}$, a one-sided test on the sample mean.

(b) Choose c for level $\alpha_0 = 0.05$. Under $H_0 : \theta = 0$, $\overline{X}_n \sim N(0, 1/n) = N(0, 1/25)$, so $\sqrt{n}\,\overline{X}_n = 5\overline{X}_n \sim N(0, 1)$. We want

$$P_{\theta=0}(\overline{X}_n > c) = 0.05 \iff P(Z > 5c) = 0.05 \iff 5c = z_{0.05} = 1.645.$$

$$\boxed{c = 1.645/5 = 0.329.}$$

(c) Type I and Type II errors.

Type I: $\alpha = P_{\theta=0}(\overline{X}_n > 0.329) = P(Z > 1.645) = 0.05 = \alpha_0$. ✓

Type II: Under $H_1 : \theta = 0.5$, $\overline{X}_n \sim N(0.5, 1/25)$, so $5(\overline{X}_n - 0.5) \sim N(0, 1)$.

$$\beta = P_{\theta=0.5}(\overline{X}_n \leq 0.329) = P\left(Z \leq \frac{0.329 - 0.5}{1/5}\right) = P(Z \leq -0.855) = \Phi(-0.855) = 1 - \Phi(0.855).$$

Using $\Phi(0.855) \approx 0.80$:

$$\boxed{\beta \approx 1 - 0.80 = 0.20.}$$

The power of the test at $\theta = 0.5$ is $1 - \beta \approx 0.80$.

(d) UMP for the composite alternative. The family $\{N(\theta, 1)\}$ has monotone likelihood ratio in $T(\mathbf{X}) = \overline{X}_n$: for any $\theta_1 > \theta_0$,

$$\frac{f_{\theta_1}(\mathbf{x})}{f_{\theta_0}(\mathbf{x})} = \exp\Big((\theta_1 - \theta_0) \sum x_i + \text{const}\Big)$$

is monotone increasing in $\sum x_i$ (equivalently in $\overline{X}_n$). By the Karlin-Rubin theorem (or DeGroot's MLR / UMP theorem), the test "reject if $\overline{X}_n > c$" with c chosen so that $P_{\theta=0}(\overline{X}_n > c) = 0.05$ is $\boxed{\text{UMP at level 0.05 for } H_0 : \theta \le 0 \text{ vs } H_1 : \theta > 0}$. The size is achieved at the boundary $\theta = 0$ since the power function is monotone increasing in θ.

(e) Smallest n^* for power ≥ 0.95 at $\theta = 0.5$. For sample size n, the level-0.05 test rejects when $\sqrt{n}\,\overline{X}_n > z_{0.05} = 1.645$, i.e. $\overline{X}_n > 1.645/\sqrt{n}$. Power at $\theta = 0.5$:

$$\pi(0.5) = P_{0.5}\big(\overline{X}_n > 1.645/\sqrt{n}\big) = P\left(Z > \frac{1.645/\sqrt{n} - 0.5}{1/\sqrt{n}}\right) = P\big(Z > 1.645 - 0.5\sqrt{n}\big)\,.$$

Require $\pi(0.5) \ge 0.95$, i.e. $1.645 - 0.5\sqrt{n} \le -z_{0.05} = -1.645$ (since $P(Z > -1.645) = 0.95$). Solving:

$$0.5\sqrt{n} \ge 3.29 \implies \sqrt{n} \ge 6.58 \implies n \ge 43.30.$$

$$\boxed{n^* = 44.}$$

(In symbolic form: $n^* = \lceil ((z_{0.05} + z_{0.05})/0.5)^2 \rceil = \lceil (2 \cdot 1.645/0.5)^2 \rceil = \lceil 43.30 \rceil = 44.$)

Question 8 (20 points) Two-Sample t-Test, CI, and F-Test

Given: $m = n = 10$, $\overline{X}_m = 18.0$, $S_X^2 = 36.0$, $\overline{Y}_n = 15.5$, $S_Y^2 = 28.0$.

(a) Pooled variance.

$$s_p^2 = \frac{S_X^2 + S_Y^2}{m + n - 2} = \frac{36.0 + 28.0}{18} = \frac{64}{18} = \frac{32}{9} \approx 3.556.$$

Hence $s_p \approx 1.886$. $\boxed{s_p^2 = 32/9 \approx 3.556}$.

(b) Two-sample t-test. The test statistic is

$$U = \frac{\overline{X}_m - \overline{Y}_n}{s_p\sqrt{1/m + 1/n}} = \frac{18.0 - 15.5}{\sqrt{32/9} \cdot \sqrt{2/10}} = \frac{2.5}{\sqrt{32/9} \cdot \sqrt{1/5}} = \frac{2.5}{\sqrt{32/45}}.$$

Compute the denominator: $32/45 \approx 0.7111$, so $\sqrt{32/45} \approx 0.8433$. Thus

$$U \approx \frac{2.5}{0.8433} \approx 2.964.$$

Under $H_0 : \mu_X = \mu_Y$, $U \sim t_{m+n-2} = t_{18}$. Reject if $|U| > t_{0.025,18} = 2.101$.

Since $|U| \approx 2.964 > 2.101$, $\boxed{\text{reject } H_0}$: there is evidence at the 5% level that $\mu_X \ne \mu_Y$.

(c) 95% CI for $\mu_X - \mu_Y$. Using the same pivot:

$$(\overline{X}_m - \overline{Y}_n) \pm t_{0.025,18} \cdot s_p\sqrt{1/m + 1/n} = 2.5 \pm 2.101 \cdot 0.8433 \approx 2.5 \pm 1.772.$$

$$\boxed{95\% \text{ CI: } (0.728,\ 4.272).}$$

Since 0 lies outside this CI, by test/CI duality the test in (b) rejects $H_0 : \mu_X - \mu_Y = 0$, consistent with the conclusion. ✓

(d) F-test for equal variances. The unbiased sample variances are

$$s_X^2 = \frac{S_X^2}{m-1} = \frac{36}{9} = 4.0, \qquad s_Y^2 = \frac{S_Y^2}{n-1} = \frac{28}{9} \approx 3.111.$$

The test statistic is

$$V = \frac{s_X^2}{s_Y^2} = \frac{4.0}{28/9} = \frac{36}{28} = \frac{9}{7} \approx 1.286.$$

Under $H_0 : \sigma_X^2 = \sigma_Y^2$, $V \sim F_{m-1,n-1} = F_{9,9}$.

For a two-sided level-0.10 test, reject if $V \leq F_{0.95,9,9} \approx 0.31$ or $V \geq F_{0.05,9,9} = 3.18$.

Since $0.31 < 1.286 < 3.18$, $\boxed{\text{fail to reject } H_0}$: there is no evidence at the 10% level that the variances differ. (This retroactively supports the equal-variance assumption used in part (b).)

(e) Wilks' theorem for the test in (b). The full parameter is $\theta = (\mu_X, \mu_Y, \sigma^2) \in \mathbb{R}^2 \times \mathbb{R}_{>0}$ (3-dim). The null $H_0 : \mu_X = \mu_Y$ imposes one linear constraint, leaving the constrained space 2-dim. By Wilks' theorem,

$$\boxed{-2 \log \Lambda(\mathbf{X}) \xrightarrow{d} \chi_1^2 \text{ under } H_0,}$$

with d.f. $= \dim \Omega - \dim \Omega_0 = 3 - 2 = 1$, matching the one constraint imposed.

End of solutions. Total: 100 points.